

Responsible AI Assignment-1

Results Report

This report covers three of the four assignment questions: color bias in MNIST, background bias in a cat/dog classifier, and class imbalance on a long-tailed CIFAR-10 subset. All three experiments follow the same basic structure: a model is trained on data containing a built-in shortcut (color, background, or class frequency), then evaluated under conditions that preserve, remove, or reverse that shortcut. The goal is to determine what each model actually learned, as distinct from what it happened to latch onto during training.

Q1 Color-Biased MNIST

Setup

MNIST digits were recolored so that digits 0–4 appear predominantly red and digits 5–9 appear predominantly blue, using a 90/10 split: 10% of images follow the opposite color rule, so the correlation is imperfect by design. A CNN and a ViT were trained on these colored digits, and the same two architectures were trained again on the original grayscale images, which contain no color information. All four models were then evaluated under three conditions: the same 90/10 rule used in training, a randomized color with no relation to the label, and a fully reversed color rule (0–4 now blue, 5–9 now red).

Results

Model	Same Bias (0.9)	Random Color	Reversed Color
CNN (color)	98.9%	97.8%	97.0%
ViT (color)	97.3%	93.3%	93.2%
CNN (grayscale)	99.1%	99.1%	99.1%
ViT (grayscale)	97.6%	97.6%	97.6%

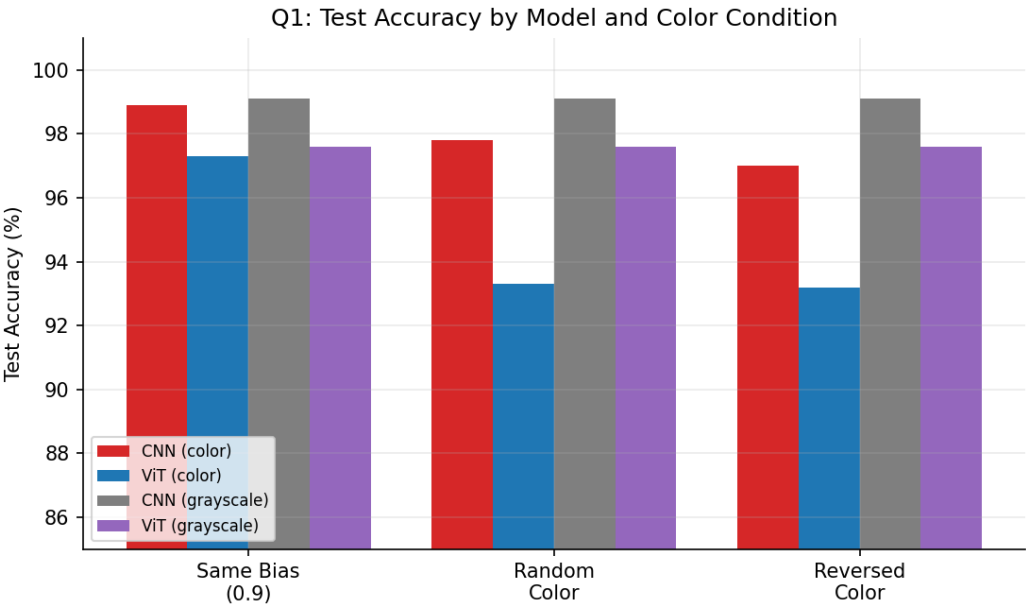


Figure 1: Test accuracy by model and color condition.

Shape vs. color diagnostic

Model	Correct Color	Reversed Color	Drop	Verdict
CNN (color)	98.9%	97.0%	1.9 pp	mostly reading shape
ViT (color)	97.3%	93.2%	4.1 pp	mostly reading shape

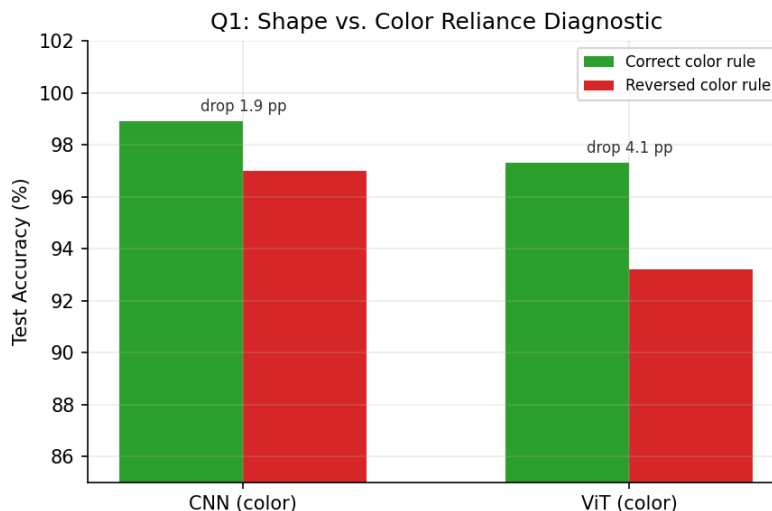


Figure 2: Accuracy under the correct vs. reversed color rule, color-trained models only.

Interpretation

All four models perform well under this setup. The two color-trained models show only a small change in accuracy when the color rule is reversed: the CNN drops by less than two points (98.9% to 97.0%), and the ViT drops by about four points (97.3% to 93.2%). If either model relied primarily on color, reversing the rule should have pushed accuracy toward chance level, and neither model approaches that. This indicates that both architectures are relying mainly on digit shape rather than color.

The grayscale models are flat across all three conditions by construction, since they never receive color information, and their accuracy is close to that of the color-trained models (99.1% versus 98.9% for CNN, 97.6% versus 97.3% for ViT). This suggests the color channel did not provide meaningful additional signal, and may have introduced a mild distraction, since neither color-trained model outperforms its grayscale counterpart.

One pattern worth noting is that the ViT shows a larger drop than the CNN under the reversed rule (4.1 percentage points versus 1.9). The difference is small, but it is consistent with the architectural distinction between the two models: the CNN’s convolution and pooling operations impose a spatial bias toward shape, while the ViT has fewer built-in assumptions and may rely somewhat more on whatever correlated feature is available, including color.

Q2 Does the Background Matter?

Setup

This experiment uses a binary cat/dog classifier built from CIFAR-10. Each 32×32 animal image is placed on a 64×64 canvas with a solid-color border, so the background accounts for roughly 61% of the image area – a substantial region the model could exploit instead of examining the animal itself. During training, 90% of cat images had a green background and 90% of dog images had a blue background. A CNN was trained from scratch and a ResNet18 pretrained on ImageNet was fine-tuned, both on this biased dataset. Both models were then evaluated on an IID test set (same 90/10 rule as training), a balanced test set (background independent of class), and a counterfactual test set (the rule reversed: predominantly blue cats and green dogs).

Results

Model	Test Set	Overall	green-cat	blue-cat	green-dog	blue-dog
CNN	IID	91.1%	99.0%	12.4%	19.5%	99.3%
CNN	Balanced	58.9%	99.6%	13.6%	23.0%	99.2%
CNN	Counterfactual	28.1%	100.0%	16.5%	24.6%	99.0%
ResNet18	IID	91.6%	97.8%	24.8%	42.5%	98.0%
ResNet18	Balanced	66.1%	98.2%	27.1%	41.1%	98.2%
ResNet18	Counterfactual	39.8%	96.7%	26.4%	40.6%	100.0%

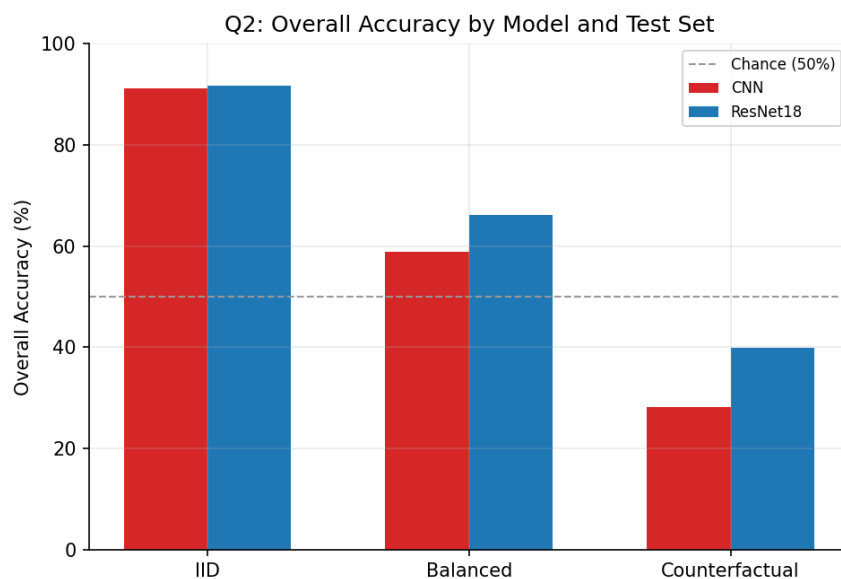


Figure 3: Overall accuracy drops hard once the background stops being a reliable cue, for both models.

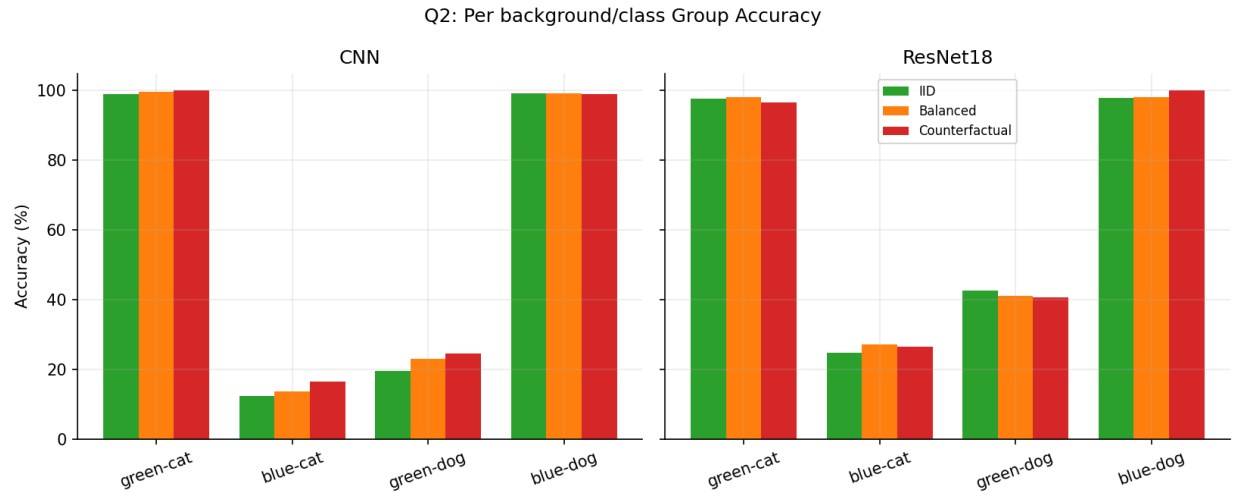


Figure 4: Per background/class group accuracy. The gap between the two “correct-correlation” groups (green-cat, blue-dog) and the two “wrong-correlation” groups (blue-cat, green-dog) is large and does not vary meaningfully across test sets.

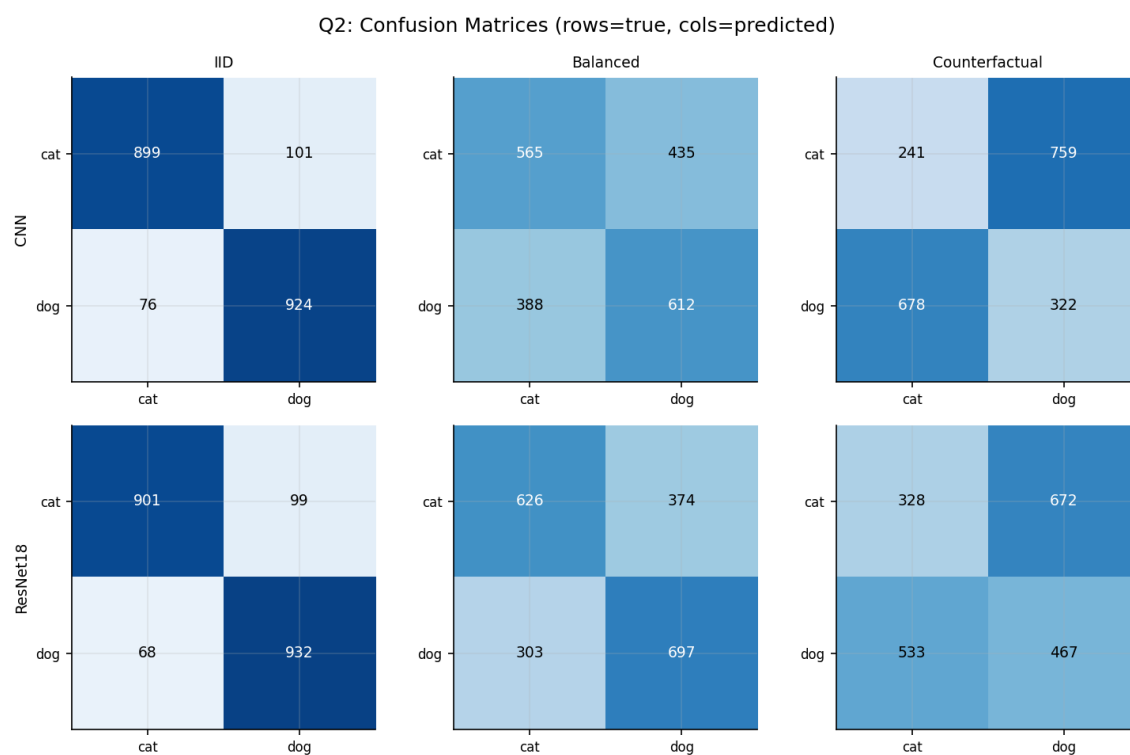


Figure 5: Confusion matrices (rows = true class, columns = predicted class).

Interpretation

This experiment provides a much clearer case of shortcut learning than Q1. Both models degrade substantially once the background correlation is reversed: the CNN falls from 91.1% to 28.1% (a 63-point drop), and the ResNet18 falls from 91.6% to 39.8% (roughly 52 points). These drops are far larger than would be expected from a model that is genuinely attending to the animal rather than its surroundings.

The per-group results explain why. On the IID test set, animals with the expected background (green-cat, blue-dog) score 97–100%, while the same animals with the reversed background (blue-cat, green-dog) score only 12–43%. This gap persists across the balanced and counterfactual test sets as well, indicating that the model does not adjust its reliance on background depending on context; it uses the background cue consistently across all conditions.

The ResNet18 performs noticeably better than the CNN on the wrong-correlation groups (24–27% versus 12–16% on blue-cat, for example), which is consistent with its ImageNet pretraining providing some prior knowledge of cat and dog appearance. However, its performance on these groups remains low overall, indicating that fine-tuning on the biased dataset overwrote a substantial portion of that pretrained knowledge rather than building on it.

The confusion matrices support this interpretation directly: on the counterfactual test set, the CNN misclassifies 759 of 1,000 cats as dogs. This is not a marginal error rate; the model is predicting close to the opposite of the correct label whenever the background contradicts the true class.

It is also worth noting that the balanced test set, where the background is uninformative rather than actively misleading, still produces a substantial accuracy drop for both models (58.9% for CNN, 66.1% for ResNet18). This occurs because in a 50/50 split, the background points to the wrong class roughly half the time, and both models continue to rely on it regardless.

Q3 The Long Tail Problem

Setup

A 7-class subset of CIFAR-10 (airplane, automobile, bird, cat, deer, dog, frog) was constructed with a heavily skewed training distribution, ranging from 5,000 airplane images down to 25 frog images – a 200:1 imbalance ratio. The test set remains perfectly balanced at 1,000 images per class, so there is no ambiguity in what constitutes strong performance. A CNN and a ViT were each trained under four conditions, using identical architecture, optimizer, and number of epochs: standard cross-entropy as a baseline, class-weighted cross-entropy, oversampling of rare classes via a weighted sampler, and focal loss with $\gamma = 2$.

Results – CNN

Class	N train	Baseline	Weighted CE	Oversampling	Focal Loss
airplane	5000	90.2%	83.5%	89.4%	91.9%
automobile	2500	92.5%	86.6%	88.3%	94.5%
bird	1250	77.2%	46.6%	79.1%	71.5%
cat	500	65.9%	72.9%	50.2%	46.8%
deer	250	41.4%	68.1%	34.5%	46.7%
dog	50	6.1%	5.2%	4.2%	7.6%
frog	25	3.6%	9.4%	2.4%	7.3%
Balanced Acc	–	53.8%	53.2%	49.7%	52.3%
Worst Class	–	3.6%	5.2%	2.4%	7.3%

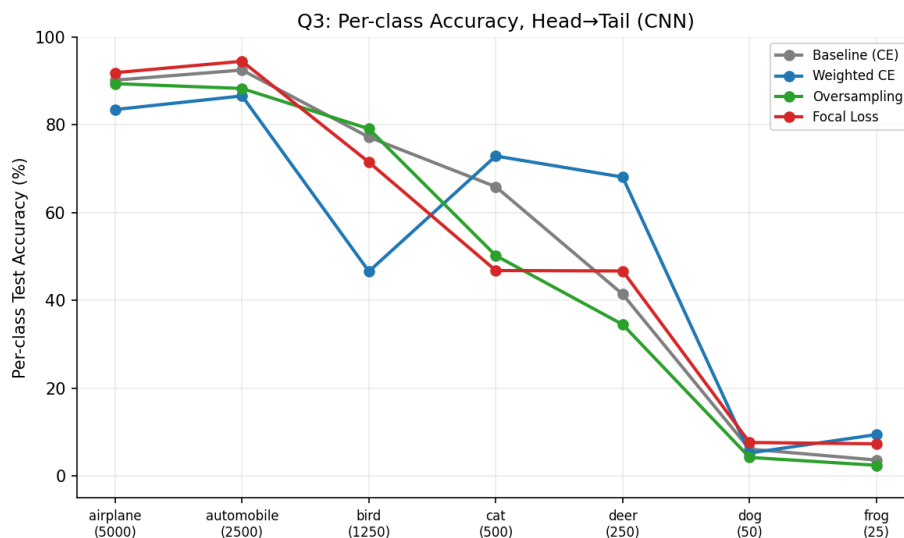


Figure 6: CNN per-class accuracy, head to tail, across all four training conditions.

Results – ViT

Class	N train	Baseline	Weighted CE	Oversampling	Focal Loss
airplane	5000	89.9%	73.0%	71.4%	87.8%
automobile	2500	83.5%	74.5%	93.8%	86.6%
bird	1250	63.2%	24.8%	46.7%	47.0%
cat	500	46.0%	43.7%	29.8%	43.8%
deer	250	21.7%	42.5%	41.1%	44.6%
dog	50	0.6%	11.5%	9.7%	3.0%
frog	25	0.1%	33.9%	2.1%	2.1%
Balanced Acc	–	43.6%	43.4%	42.1%	45.0%
Worst Class	–	0.1%	11.5%	2.1%	2.1%

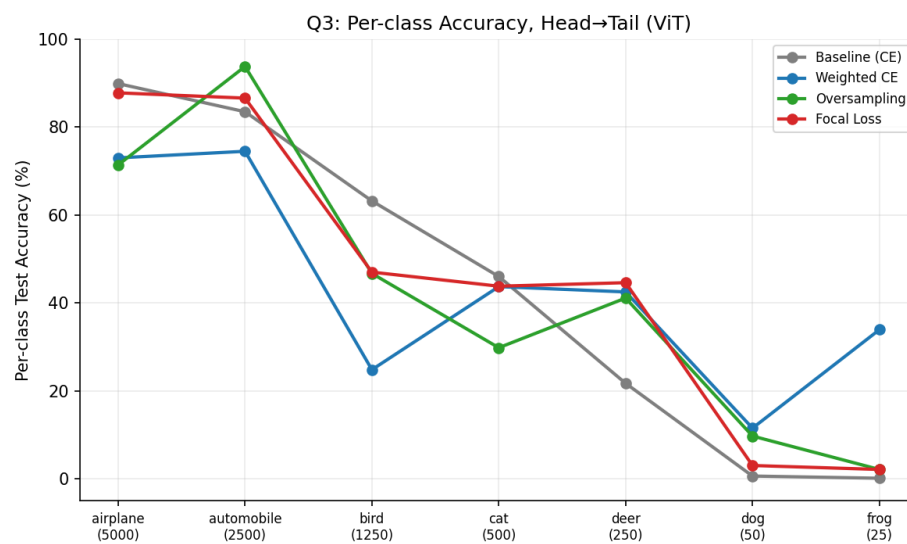


Figure 7: ViT per-class accuracy, head to tail, across all four training conditions.



Figure 8: Balanced accuracy and worst-class accuracy by intervention and model. Since the test set is perfectly balanced, Overall Accuracy equals Balanced Accuracy in every run, so it is omitted here.

Interpretation

The baseline collapse on tail classes is consistent with what a 200:1 imbalance ratio would predict. With only 25 training images, frog accuracy is 3.6% for the CNN and 0.1% for the ViT; the ViT figure is close enough to zero that the model can be considered to have failed on this class entirely.

A notable finding is how misleading the balanced-accuracy metric is when reported on its own. The baseline CNN achieves 53.8% balanced accuracy, which appears reasonable in isolation, but this figure is driven almost entirely by the airplane and automobile classes (both above 90%), while dog and frog accuracy remain in the single digits. This distinction is not visible from overall accuracy alone; it requires a per-class breakdown.

None of the three interventions resolve the problem cleanly, and their effects differ substantially between the two architectures. Weighted cross-entropy produces a large improvement in worst-class accuracy for the ViT (0.1% to 11.5%) but only a marginal change for the CNN (3.6% to 5.2%), and it reduces head-class accuracy for both models – the CNN’s airplane accuracy, for example, falls from 90.2% to 83.5%. Oversampling performs worst overall and slightly reduces the CNN’s worst-class accuracy (3.6% to 2.4%), likely because repeatedly sampling the same 25 frog images encourages memorization rather than generalization. Focal loss yields the best result for the CNN (3.6% to 7.3%) while largely preserving head-class accuracy, and produces a small, non-harmful improvement for the ViT (0.1% to 2.1%).

Even the strongest results obtained here – 7.3% for the CNN and 11.5% for the ViT – remain far below the 90%+ accuracy achieved on head classes. The interventions provide modest improvement but do not come close to resolving tail-class collapse under this level of imbalance.

This result speaks directly to the question posed in the assignment: whether a model that achieves high overall accuracy but consistently fails on rare classes should be considered reliable. Based on these results, it should not be. A model that fails consistently on tail classes is unsuitable for deployment in any setting where those classes are relevant, regardless of how strong its overall accuracy appears.